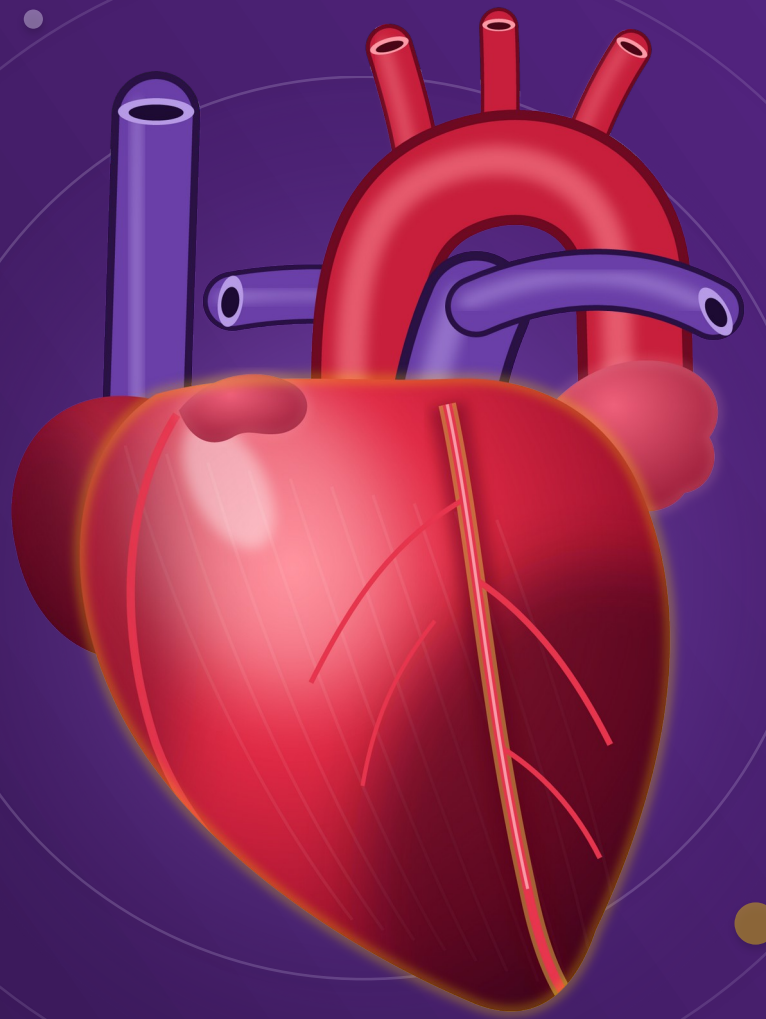
MACHINE LEARNING PROJECT

HEART DISEASE RISK PREDICTION

A Machine Learning Approach to Predicting Heart Disease



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University: University Of larkano
Subject Machine learning

Introduction & Problem Statement



A major global health problem

Cardiovascular diseases are the leading cause of death worldwide (WHO).



Early risk assessment matters

Finding higher-risk patients early helps doctors decide who needs further tests first.



Our question

Can a model learn from routine clinical measurements (age, chest-pain type, heart rate, ECG results) to estimate whether a patient is likely to have heart disease?



Prediction ≠ Diagnosis

- The model gives a risk estimate learned from historical patient records.
- A diagnosis needs a clinician and medical tests (for example, coronary angiography).
- This project is an educational machine-learning prototype.

920 records · 4 hospitals · 13 input features

Project Objectives & Workflow

What the project set out to do

1. Explore the UCI heart-disease data and understand each feature
2. Clean it: missing values, impossible zeros, encoding and scaling
3. Train and tune 6 classifiers with 5-fold cross-validation
4. Choose the best by F1, test it once and save it for predictions

Workflow



Expected output: a saved pipeline (heart_model.pkl) that labels a new patient “Healthy” or “Heart Disease Risk” with a probability. It is a learning prototype, not a clinical tool.

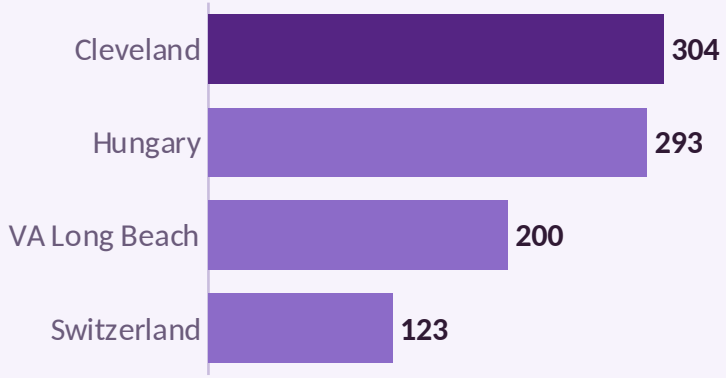
Dataset Overview

920 patient records (rows)

16 columns: 13 used as model inputs

4 hospital sources

Records by source



Numerical · 6

age (years) · trestbps (resting blood pressure, mm Hg) · chol (serum cholesterol, mg/dl) · thalch (max heart rate) · oldpeak (exercise ST depression) · ca (major vessels, 0–3)

Categorical · 5 + source

sex · cp (chest-pain type, 4) · restecg (resting ECG, 3) · slope (ST-segment slope, 3) · thal (thallium test, 3) · dataset (hospital source, 4)

True / False · 2

fbs (fasting blood sugar > 120 mg/dl) · exang (exercise-induced angina)

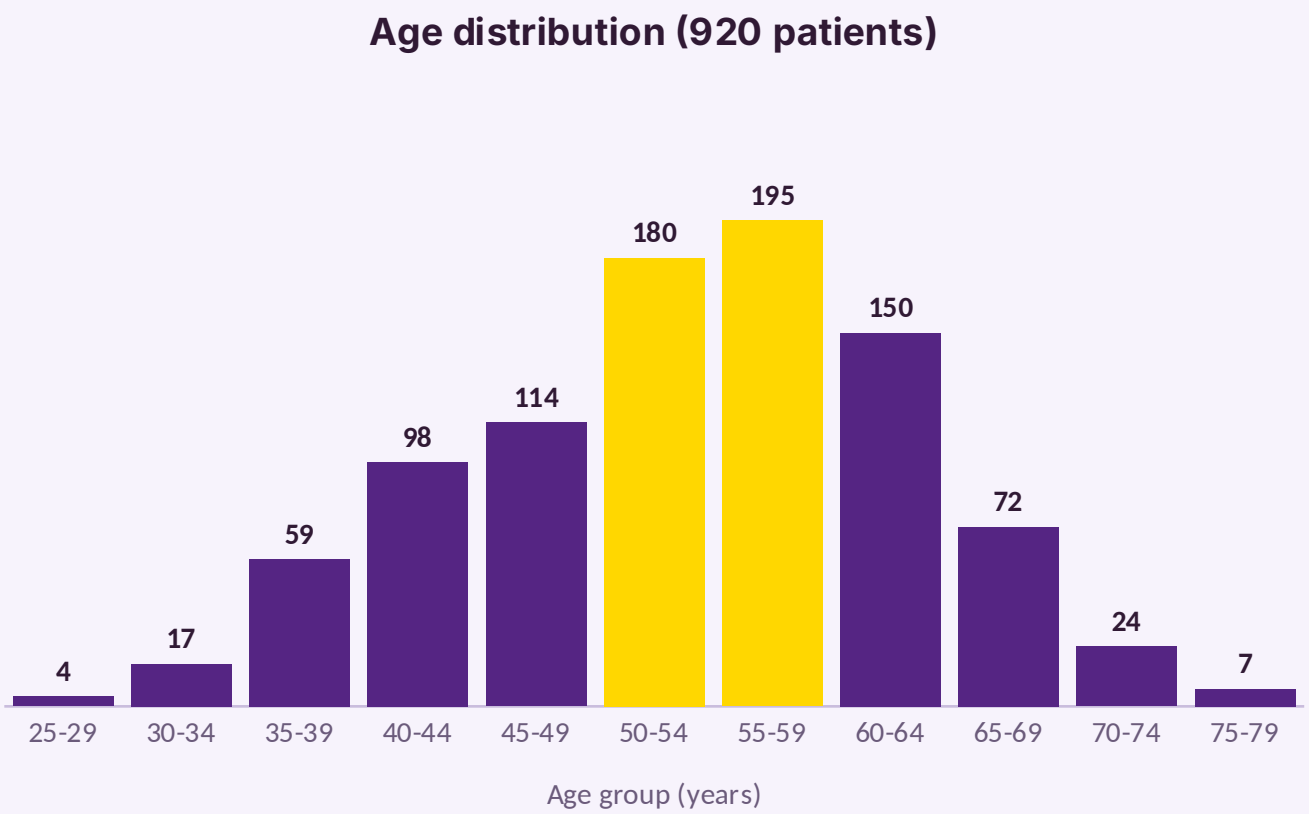
Target & ID

num: 0 = no disease, 1–4 = disease present · id: row number (not a feature)

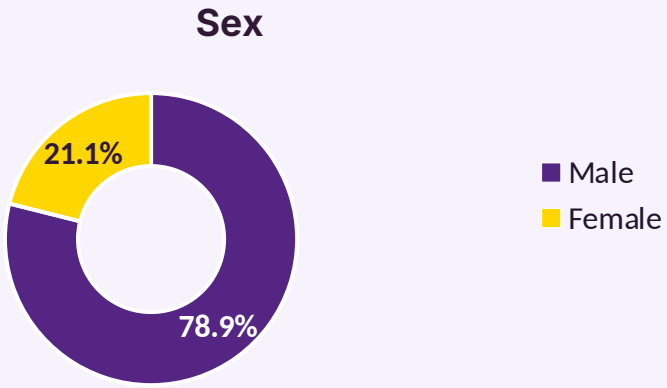
First rows of heart_disease_uci.csv (selected columns)

age	sex	cp	trestbps	chol	thalch	exang	num
63	Male	typical angina	145	233	150	False	0
67	Male	asymptomatic	160	286	108	True	2
67	Male	asymptomatic	120	229	129	True	1

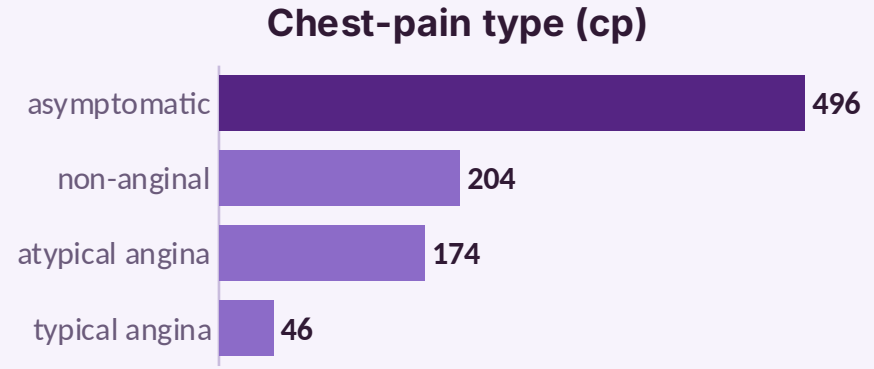
Univariate Analysis



Observation: ages 28–77 (mean 53.5, median 54); the two busiest groups are 50–59 (yellow).



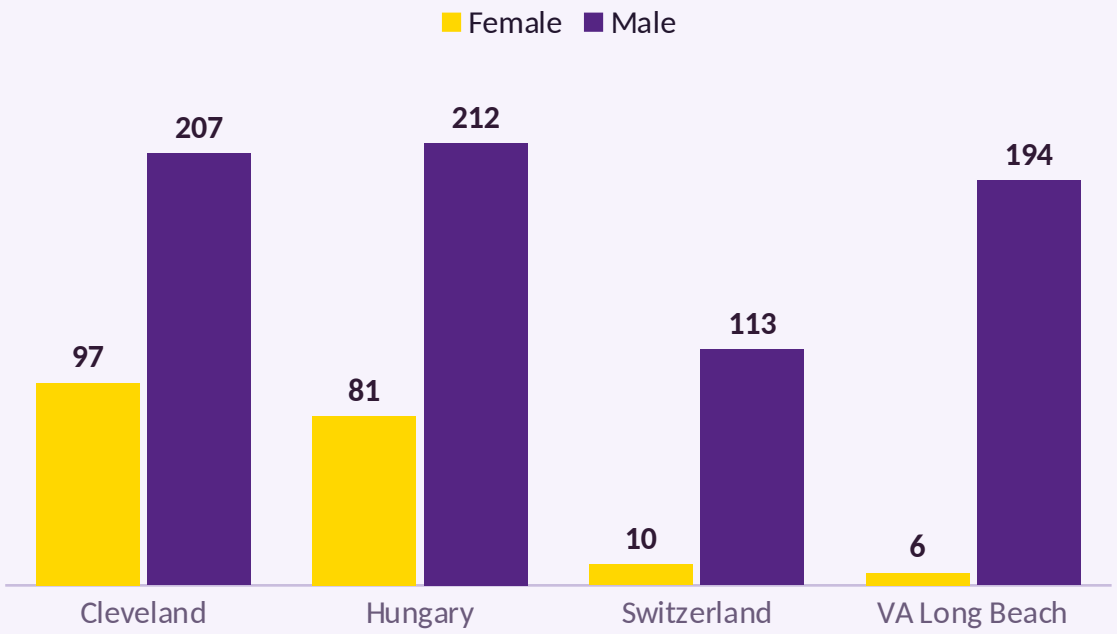
726 men vs 194 women: the data is unbalanced by sex (79% male).



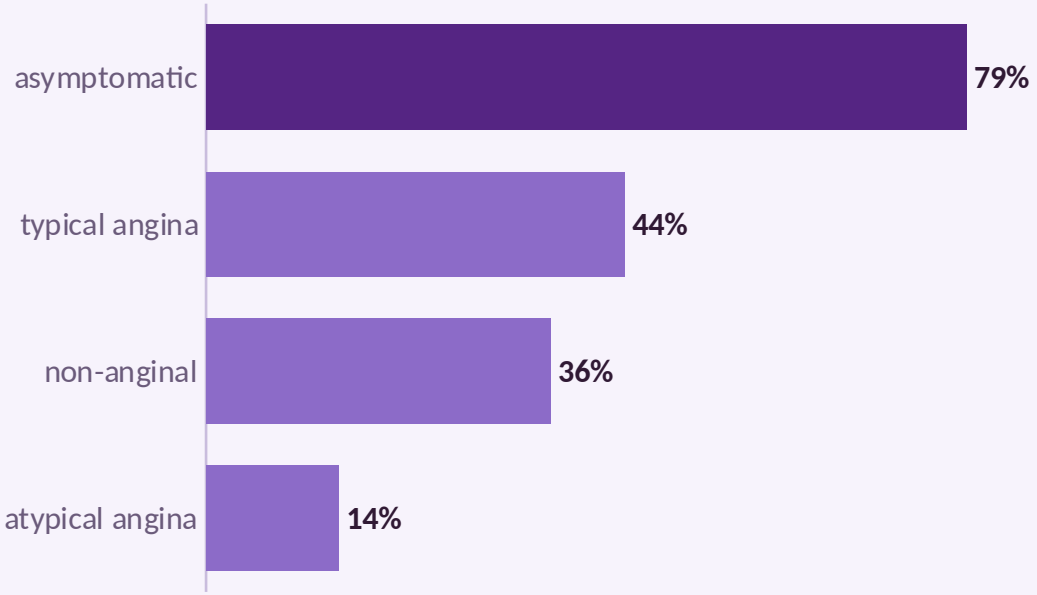
"Asymptomatic" is the most common type (496 patients, 54%).

Bivariate Analysis & Key Findings

Patients by source and sex



Heart-disease share (num > 0) by chest-pain type



Right chart: computed from the dataset for this presentation (not plotted in the notebook). Both show associations, not causes.

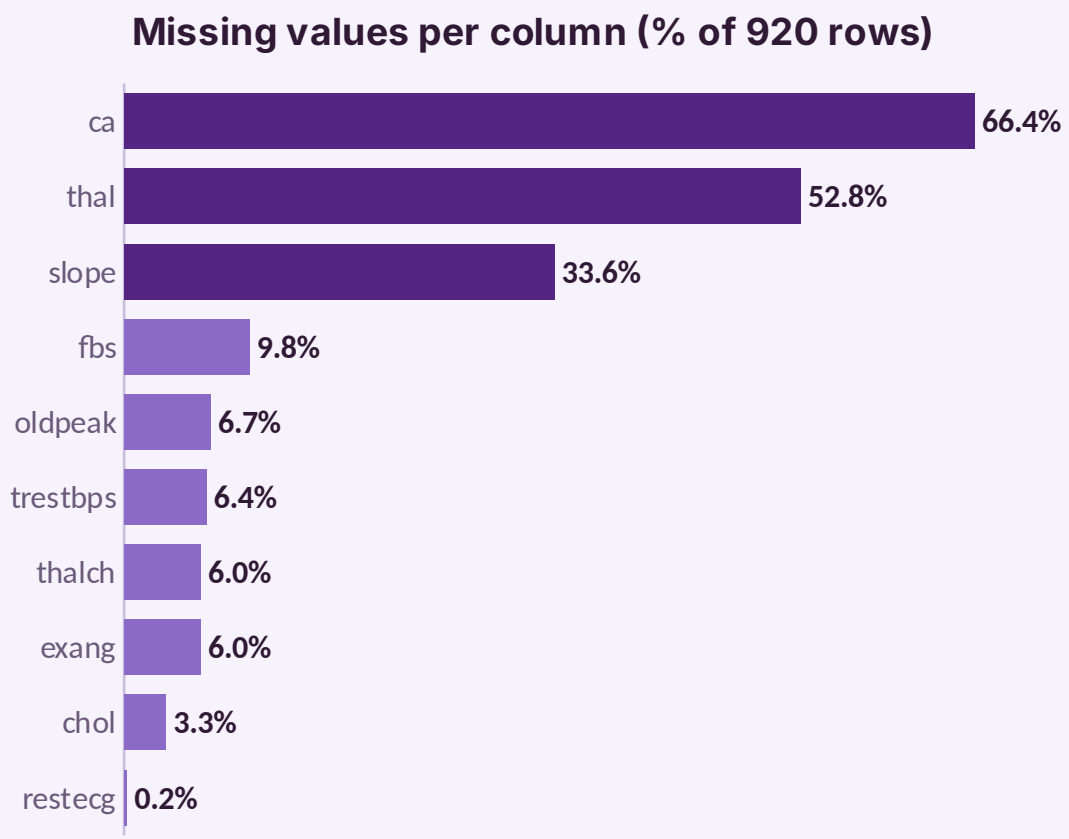
Hungary has the most male patients (212); VA Long Beach has only 6 women.

Mean age varies by source: 47.9 years (Hungary) vs 59.4 (VA Long Beach).

Disease share varies by source (Switzerland 93%, Hungary 36%), so “dataset” was dropped.

Data Quality & Missing Values

C · PREPROCESSING



1,759
missing cells before

0
missing after imputation

Numerical & true/false columns

trestbps, chol, thalch, oldpeak, ca, fbs, exang → IterativeImputer, one column at a time, so each gap gets the column mean (e.g. ca = 0.68).

Categorical columns

restecg, slope, thal → filled with the mode (the most frequent category in each column).

Inside the final pipeline

SimpleImputer: median for numbers, most frequent for categories. chol = 0 is first turned into “missing”.

Hidden missing values: 172 patients have cholesterol recorded as 0, which is impossible.

Encoding, Scaling & Outlier Handling

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Encoding

OneHotEncoder turns sex, cp, restecg, slope and thal into 0/1 columns: 13 inputs become 23 model features.

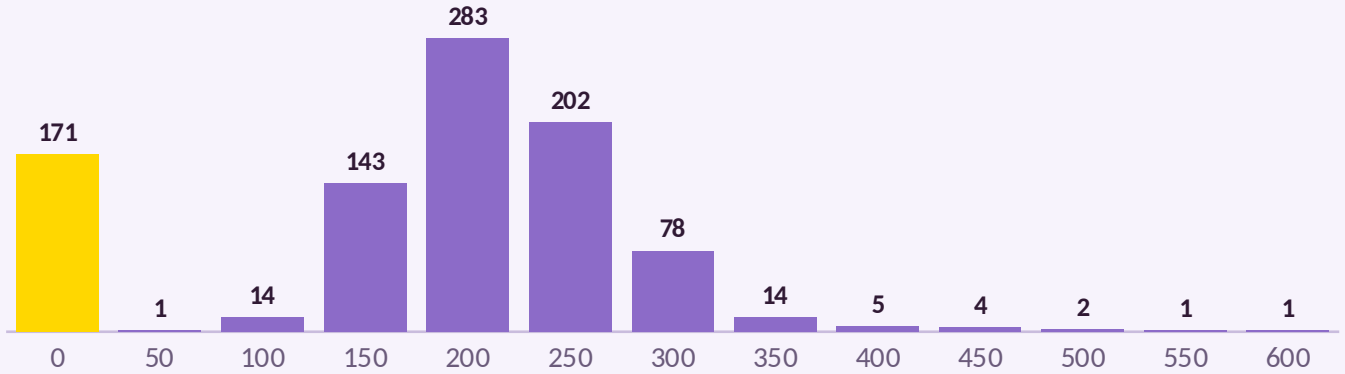
Scaling

StandardScaler: z = (x − mean) ÷ std for the 8 numeric columns, so no feature dominates because of its units.

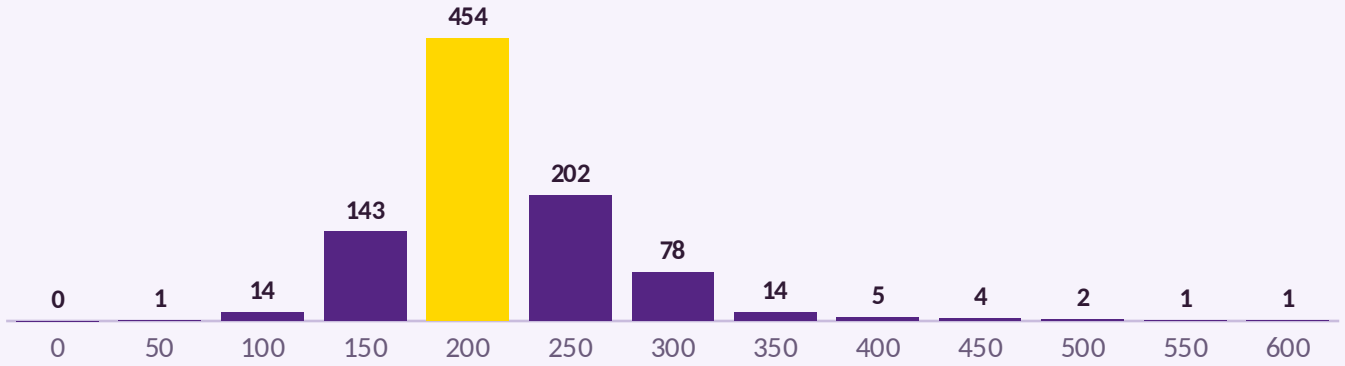
Outliers

Box plots of 5 numeric columns. Removed 1 impossible row (trestbps = 0) → 919 rows. chol = 0 → missing; other extremes kept.

Before: cholesterol as recorded (919 rows, 171 zeros)



After: zeros → missing → training median (236 mg/dl)

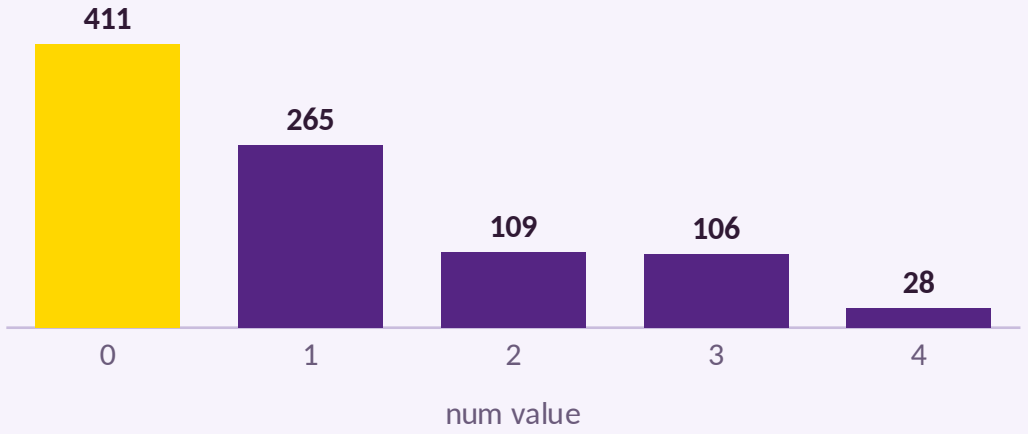


Bars: cholesterol in mg/dl, bins of 50 (label = bin start). Why preprocess? Impossible values teach wrong patterns.

Target, Features & Train/Test Split

C · PREPROCESSING

Original target num (919 patients)



411
Healthy (y = 0) · 44.7%

508
Heart disease (y = 1) · 55.3%

```
y = (df['num'] > 0).astype(int)
```

13 input features

age · sex · cp · trestbps · chol · fbs · restecg · thalch · exang · oldpeak · slope · ca · thal

Dropped: id (row number) and dataset (hospital source). No automatic feature selection.

Stratified 80 / 20 split

Training: 735 (406 disease · 329 healthy) | **Test: 184**

```
train_test_split(X, y, test_size=0.2, stratify=y,
random_state=42)  ·  test: 102 disease · 82 healthy
```

Avoiding data leakage

Good: imputer, scaler and encoder sit inside one Pipeline, so they learn only from training folds.

To fix: the earlier mean / mode imputations ran on all rows before the split.

Algorithms Used (6 Tuned Classifiers)

D · MACHINE LEARNING

Logistic Regression

Weights each feature and turns the total into a probability with the sigmoid curve.

Why here: Simple, fast, interpretable baseline.

Support Vector Machine (SVC)

Finds the boundary with the widest margin between the classes; the RBF kernel lets it curve.

Why here: Strong on small, scaled datasets.

K-Nearest Neighbors

Finds the k most similar training patients and takes a majority vote.

Why here: Intuitive; relies on scaled features.

Best CV F1

Random Forest

Builds many decision trees on random samples of rows and features, then lets them vote.

Why here: Handles mixed, non-linear data; robust.

Gradient Boosting

Adds small trees one after another, each correcting the errors of the previous ones.

Why here: Often very accurate on tabular data.

XGBoost

An optimised, regularised implementation of gradient boosting.

Why here: Fast and accurate; limits overfitting.

Decision Tree, AdaBoost, Naive Bayes and LightGBM are listed in the notebook, but only these 6 models were trained and compared.

Model Training & Tuning

D · MACHINE LEARNING

Training set

735 patients · 13 features

Pipeline

impute → scale / one-hot → model

GridSearchCV

5-fold stratified CV · scoring = F1

Best settings

72 combinations · 360 fits

Final test

best CV F1 model → 184 test patients

Core code (simplified from the notebook)

```
num = Pipeline([("imp", SimpleImputer(strategy="median")),
                ("sc", StandardScaler())])
cat = Pipeline([("imp", SimpleImputer(strategy="most_frequent")),
                ("oh", OneHotEncoder(handle_unknown="ignore"))])
prep = ColumnTransformer([("num", num, num_cols),
                          ("cat", cat, cat_cols)])
pipe = Pipeline([("prep", prep), ("m", model)])  # 6 models
cv = StratifiedKFold(5, shuffle=True, random_state=42)
gs = GridSearchCV(pipe, grid, cv=cv, scoring="f1")
gs.fit(X_train, y_train)
```

Grids searched

LR	C 0.01/0.1/1/10
SVC	C 0.1/1/10 · gamma scale/0.01/0.1
RF	trees 200/400 · depth 4/6/10/None · leaf 1/3/5
GB	trees 100/200 · rate 0.03/0.05/0.1 · depth 2/3
XGB	trees 100/200 · depth 2/3/4 · rate 0.03/0.05/0.1
KNN	k 5/7/9/11/15

RF winner: 200 trees · max depth 6 · min 1 per leaf

Model Evaluation on the Test Set

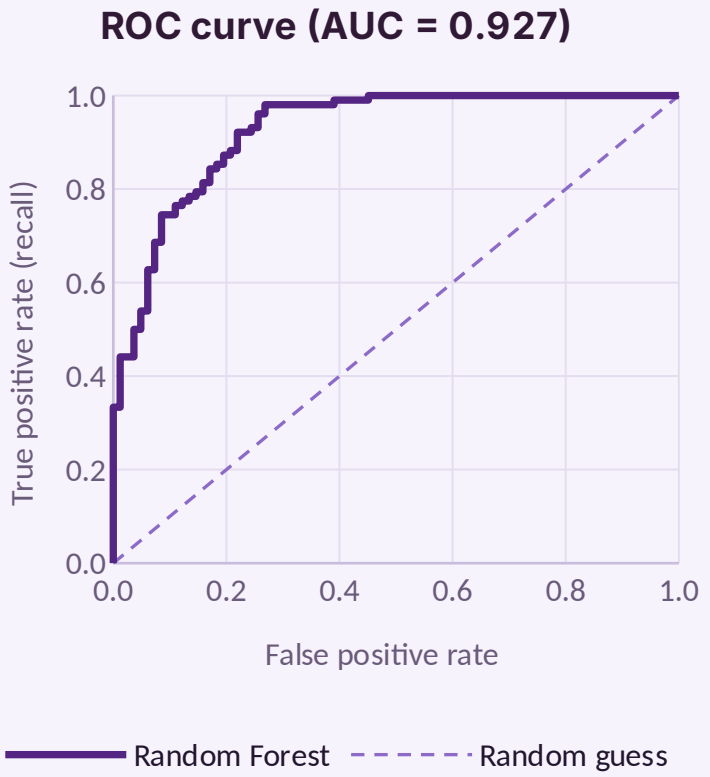
E · RESULTS & CONCLUSION

Random Forest · 184 unseen patients (20% hold-out) · decision threshold 0.5

	Predicted healthy	Predicted disease
Actual healthy	**62** TN · correctly cleared	**20** FP · false alarm
Actual disease	**10** FN · missed case	**92** TP · correctly flagged

Confusion matrix (rows = actual, columns = predicted)

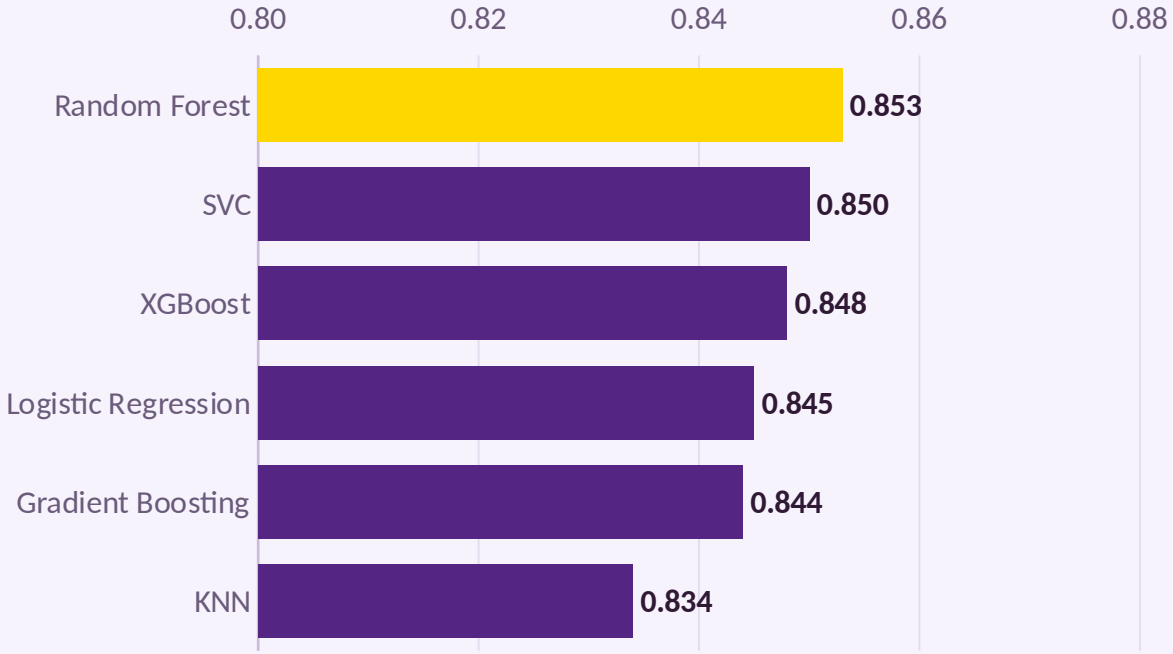
83.7%	Accuracy (154 / 184)
90.2%	Recall: disease found (92 / 102)
82.1%	Precision: flagged & ill (92 / 112)
0.86	F1-score (disease class)
0.927	ROC-AUC



⚠ **Why recall matters:** a false negative tells a sick patient they look healthy; 10 of 102 disease cases were missed. *84% accuracy on 184 patients from the same 4 hospitals is not proof of clinical reliability.*

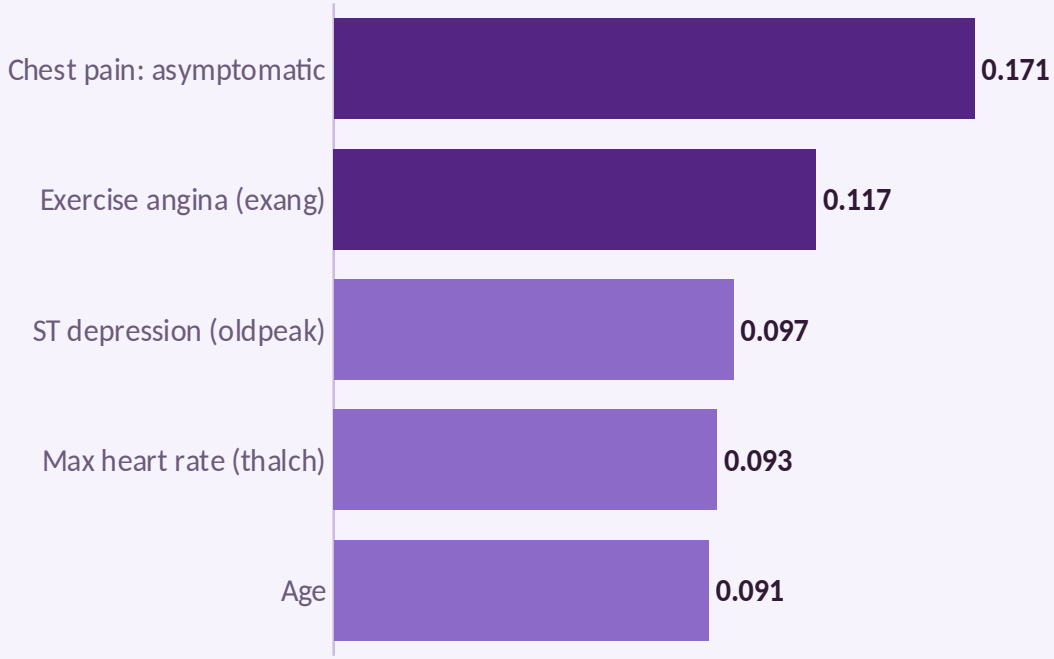
Model Comparison & Best Model

Cross-validated F1 (5-fold, training set)



Axis starts at 0.80: the six scores are close (0.834–0.853).

Top 5 features (Random Forest)



Impurity-based importance: what the model uses, not what causes disease.

Best model: Random Forest CV F1 0.853 · test accuracy 83.7% · recall 90.2% · ROC-AUC 0.927

Why F1? It balances precision and recall for the disease class; choosing on CV keeps the test set for one final check.

Prediction Demo & Limitations

E · RESULTS & CONCLUSION

Predicting a new patient (from the notebook)

Age	**55**	Max heart rate	**150**
Sex	**Male**	Exercise angina	**False**
Chest pain	**typical angina**	ST depression	**1.5**
Resting BP	**140 mm Hg**	ST slope	**flat**
Cholesterol	**230 mg/dl**	Major vessels (ca)	**1**
Blood sugar > 120	**False**	Thal	**normal**
Resting ECG	**normal**		

`model.predict_proba()`

Heart Disease Risk **52.4%**



Probability ≥ 0.5 → class 1. 52.4% sits just above the 0.5 threshold (white mark): an uncertain case, not a diagnosis. Test patient #1: 44.2% → predicted Healthy.

Limitations

- Small, old dataset: 919 patients from 4 hospitals; 79% male.
- Heavy filling-in: ca (66%) and thal (53%) were mostly missing.
- One train/test split; no validation on new hospitals.
- Probabilities are not calibrated clinical risk scores.
- Some imputation ran before the split (small data leakage).

Educational prototype: not a substitute for a medical professional.

Conclusion & What I Learned

E · RESULTS & CONCLUSION

What we found

0.853
CV F1: Random Forest, best of 6

83.7%
test accuracy (184 patients)

90.2%
recall: 10 of 102 cases missed

0.927
ROC-AUC

Objectives

- Explored and cleaned the data
- Compared 6 tuned models with cross-validation
- Evaluated once on unseen test data
- Saved the model and demonstrated a prediction
- Clinical validity: not tested (outside the project's scope)

What I learned

Clean data first
Missing values and impossible zeros must be fixed before training.

Pipelines
One pipeline keeps new data cleaned exactly like training data.

Recall matters
In medical problems, missing a sick patient is the worst error.

Fair model comparison
Cross-validation picks the best model without touching the test set.

Honest results
A model gives a risk estimate, not a medical diagnosis.

Thank You

Questions?

